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Antidepressant-like effects of albiflorin extracted from *Radix paeoniae* Alba

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ABSTRACT

Ethnopharmacology relevance: Albiflorin, a monoterpene glycoside, is a main component of *Radix paeoniae* Alba, which could be a Chinese herbal medicine used in the treatment of psychiatric disorders. However, the exact role of albiflorin in depression is poorly understood.

Aim of the study: The current study aimed to evaluate the antidepressant effect of albiflorin in mice and rats, and the possible mechanism was also determined.

Materials and methods: The antidepressant-like effects of albiflorin was determined by using animal models of depression including forced swim and tail suspension tests in mice and chronic unpredictable stress (CUS) in rats. The acting mechanism was explored by determining the effect of albiflorin on the expression of brain-derived neurotrophic factor (BDNF) in the hippocampus by western blot and the levels of monoamine in the hippocampus by HPLC.

Results: Our results showed that 7 days treatment with albiflorin significantly decreased immobility time in the forced swimming test (FST) and the tail suspension test (TST) at doses of 3.5, 7.0 and 14.0 mg/kg without alter the locomotor activity in mice. Moreover, western blot analysis showed that albiflorin could increase the expression of BDNF in the hippocampus. We further exposed rats to a chronic unpredictable stress (CUS) protocol for a period of 35 d to induce depressive-like behaviors. We found that chronic treatment with albiflorin, at doses of 7.0 and 14.0 mg (i.g., once daily for 35 d), restored the sucrose preference in CUS rats. In the open-field test, albiflorin significantly increased the number of crossings and rearings in the CUS rats at three doses. Moreover, chronic treatment with albiflorin up-regulated the hippocampal BDNF expression levels and the hippocampal 5-HT, 5-HIAA, and NA levels.

Conclusion: Albiflorin produced significant antidepressant-like effects, which were closely related to the hippocampal 5-HT/NE increase and BDNF expression. Our data indicated that albiflorin could be a potential anti-depressant drug.

1. Introduction

Mood disorders are among the most prevalent forms of mental illness. Severe forms of depression are serious emotional disorders that cause a significant burden on health worldwide. Depression should not be viewed as a single disease but rather as a heterogeneous syndrome comprising numerous diseases of distinct causes and pathophysiologies (Nestler et al., 2002). Although many synthetic chemical antidepressants can improve depressive symptoms, approximately 30% of depressive patients fail to respond satisfactorily to commercially available antidepressants due to the undesirable side effects of antidepressants, such as suicidal tendencies, sexual dysfunction, and sleep disorders (Papakostas et al., 2008). Therefore, there is strong need for research on antidepressants with less adverse effects. Currently, traditional herbs provide a prospective alternative in the treatment of depression.

Preclinical and clinical lines of evidence suggest that disturbed

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monoaminergic neurotransmission is one of most important mechanisms underlying the pathophysiology of depression. The monoamine hypothesis has demonstrated that serotonin (5-HT), noradrenaline (NE) and dopamine (DA) are the important neurotransmitters involved in the etiology of depression (Nutt, 2008). Indeed, most of the current antidepressants act on one or more mechanisms compatible with the monoamine hypothesis, such as inhibition of the reuptake of 5-HT and/or NE. A large amount of evidence from various studies has indicated that the levels of monoamine neurotransmitters in the brain increased compared with that of controls after antidepressant treatments (Elhwuegi, 2004). Brain-derived neurotrophic factor (BDNF), a type of nerve growth factor, is predominantly expressed in the central nervous system and related to the survival and maintenance of neuronal function. Elevating the expression of BDNF plays a crucial role in the treatment of depression (Haghighi et al., 2013). These effects were similar to a classic antidepressant, which can regulate the monoamine neurotransmitters in the brain. For example, BDNF levels can be restored using chronic venlafaxine treatment (Xu et al., 2004). Similarly, the chronic social defeat stress protocol with a prolonged administration with imipramine normalizes behavioral alterations while restoring decreased levels of neurotrophin (Tsankova et al., 2006). In human samples, Chen et al. first demonstrated that antidepressant treatment, such as SSRIS, venlaflaxine (Dowlatshahi et al., 1999), may normalize hippocampal levels of BDNF(Chen et al., 2001).

Albiflorin, a monoterpene glycoside, is a main component of *Radix paeoniae* Alba, which is a Chinese herbal medicine often used in the treatment of depression-like disorders. *Radix paeoniae* Alba, commonly known as the peony, is the Ranunculaceae peony root of *Paeonia lactiflora* Pall, which is often used in Chinese herbal formulas, including Sini-San and Xiaoyao-San, for the treatment of depression-like disorders (Chen et al., 2008; Mao et al., 2008, 2012). Among these components, the monoterpene glycosides extracts, such as paeoniflorin and albiflorin, are usually described as the most important active components of the peony (He et al., 2010). Reports have shown that paeoniflorin has antidepressant effects in rats and mice (Qiu et al., 2013a, 2013b). However, the antidepressant effect of albiflorin, the other important component of the monoterpene glycosides extract, has not been reported. First, we chose two classic animal behavior despair tests, the forced swimming test (FST) and tail suspension test (TST), to evaluate the antidepressant-like activity of albiflorin in mice. We then used a chronic unpredictable stress model to further evaluate the antidepressant-like effects of albiflorin, and the hippocampal monoamine levels and BDNF expression were also measured.

2. Materials and methods

2.1. Animals and housing

Male ICR mice (18–22 g) and Wistar rats (140–160 g) were purchased from SPF (Beijing) Laboratory Animal Technology Co., Ltd. All of the mice or rats were caged in groups (five per cage) and allowed to acclimate for 3 d prior to experimental testing. The animals were housed in a temperature (22 ± 2 °C)- and humidity (50 ± 10%)-controlled room. All animals were maintained on a 12 h/12 h light–dark cycle with the lights on at 8:00 a.m. Food and water were available ad libitum in the home cages unless otherwise specified. All of the experimental procedures were performed in accordance with the National Institutes of Health (NIH) Principles of Laboratory Animal Care (publication No. 80-23, revised 1996).

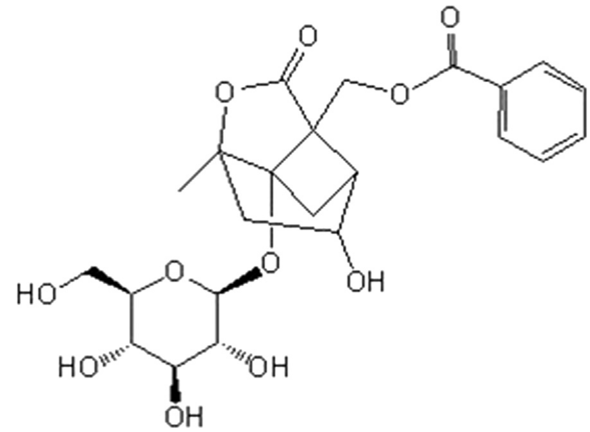

Fig. 1. The chemical structure of albiflorin.

2.2. Chemicals

Albiflorin (purity 98.63% according to high-performance liquid chromatography (HPLC)) was obtained from Beijing Wonner Biotech Co., Ltd. (China). The molecular formula is $C_{23}H_{28}O_{11}$, and the chemical structure is shown as Fig. 1. Fluoxetine hydrochloride (purity ≥ 99% according to HPLC) was purchased from the Changzhou Fourth Pharmaceutical Factory (China).

2.3. Antidepressant-like effects of a subchronic treatment with albiflorin in mice

2.3.1. Drugs and administration

Mice were randomly divided into five groups ($n = 15$ per group) for the forced swimming test (FST), tail suspension test (TST) and locomotor test. The albiflorin (3.5, 7.0 and 14.0 mg/kg), fluoxetine (10.0 mg/kg) or distilled water were given orally daily for 7 days. The doses of albiflorin were relied on preliminary experiments. On the last day, the drugs were given 1 h prior to the test. After the test, the brains were rapidly removed, and the bilateral hippocampus was isolated from the brain after a rapid dissection of the cranium. The tissues were rapidly frozen and stored at −80 °C until they were processed for biochemical estimations.

2.3.2. Forced swimming test (FST)

The FST is widely used for detecting antidepressant activity and is based on Porsolt et al. (1977) with slight modifications. Briefly, mice were placed in an inescapable plexiglass cylinder (20 cm in height × 10 cm in diameter) filled with water to a depth of 15 cm maintained at 24 °C for 6 min. The duration of immobility (seconds) was defined during the last 4 min period of the test that the mouse spent in the water without struggling and made no further attempts to escape. An elevation in the duration of immobility was considered to be a criterion of an antidepressant effect (Cryan and Holmes, 2005; Gao et al., 2013).

2.3.3. Tail suspension test (TST)

The TST was based on the method of Steru et al. (1985) with slight modifications. The animals were individually suspended 15 cm above the floor using adhesive tape that was placed approximately 1 cm from the tip of the tail for 6 min. The duration of immobility (seconds) was measured during the last 4 min of the test. The mice were considered to be immobile only when they hung passively and completely motionless.

2.3.4. Locomotor activity

Mice were used for the locomotor activity experiment, and the experiment was performed between 8:00 a.m. and 12:00 a.m. The locomotor activity was measured automatically using the SuperState v3.0 system (AniLab Software & Instruments Co., Ningbo, China), which consisted of nine sound proof boxes with dim illumination and a fan in each box. Horizontal locomotor activity was recorded through infrared beams. The mouse was monitored under infrared light. The locomotor activity was recorded for 10 min and the distance traveled (cm) was calculated.

2.4. TableChronic unpredictable stress (CUS) protocol

Rats were subjected to chronic unpredictable stress (CUS) for five weeks. The CUS procedure was performed as previously described (Gronli et al., 2005; Papp et al., 1996) with minor modifications. The CUS protocol consisted of various stressors: (1) food deprivation for 24 h, (2) water deprivation for 24 h, (3) exposure to stroboflash and white noise for 2 h, (4) cage tilt (45°) for 12 h, (5) overnight illumination, (6) soiled cage for 24 h, (7) forced swimming at 4 °C for 6 min, (8) restraint for 2 h, and (9) tail pinch for 6 min. The stressors were administered in a semi-random manner at any time of the day, once or twice per day. The stress sequence was changed every week to make the stress procedure unpredictable. These stressors were randomly scheduled over a one-week period and were repeated throughout the five-week experiment. Control animals were housed in a separate room and had no contact with the stressed animals. The experimental procedure is shown in Fig. 2. On day 35 and 37, the sucrose preference and open-field test were performed after 1 h of drug administration, respectively. After the behavioral test, the rats were sacrificed and their brains were rapidly isolated to obtain the bilateral hippocampus. The tissues were rapidly frozen and stored at −80 °C until they were processed for biochemical estimations.

2.4.1. Sucrose preference test

The sucrose preference test was performed as previously described (Willner et al., 1987) with minor modifications. This procedure included training and testing courses. After 3 days of acclimatization, before the CUS test, rats were trained to consume a sucrose solution (1%, w/v). In the training course during 48 h of food and water deprivation, we put two bottles of sucrose solution into each cage for 24 h and then replaced one bottle of sucrose solution with water for 24 h. Three days later, after 23 h of water and food deprivation, a 1 h baseline test was performed, in which the rats could choose from two pre-weighted bottles, one with a 1% (w/v) sucrose solution and the other with water. The sucrose preference (SP) was calculated using the following formula: SP = sucrose intake (g)/{sucrose intake(g) + water intake (g)} × 100%.

Then, we divided the rats into six parallel groups according to the SP baseline data: control (no stressor), stress-control, stress-fluoxetine (10 mg/kg, i.g.), and stress-albiflorin (3.5, 7.0 and 14.0 mg/kg, i.g.). Each group included 10 rats. All of the treatments were administered by oral gavage at 2 mL/kg body weight once a day between 8:00 and 9:00 a.m. from the beginning of the stress protocol. On day 35, the sucrose preference test was performed again to estimate the CUS model and drug effects.

2.4.2. Open-field test

The (Kulkarni and Dandiya, 1973) open-field apparatus consisted of a 90 cm × 90 cm gray wooden box with 45 cm-high boundary walls. This apparatus was divided into 36 equal squares by red lines on the floor. Each animal was placed in the central square and was observed for 5 min. After each trial, the apparatus was cleansed with 5% ethanol. The number of crossings and rearings was recorded.

2.5. Western blotting analysis

After the subchronic treatment and chronic unpredictable stress protocol, bilateral hippocampi of mice and rats were rapidly removed and the hippocampus was lysed in RIPA buffer containing phosphatase inhibitors and protease inhibitors. The protein concentration was quantitated by BCA assay (Pierce, Rockford, IL, USA). A sample consisting of 50 μg of protein was loaded into each lane of polyacrylamide gels. Protein lysates were subjected to electrophoresis in 15% SDS-PAGE gels and were transferred to polyvinylidene difluoride (PVDF) membranes. The membranes were blocked with 5% BSA for 2 h and incubated with rabbit anti-BDNF (Sigma-Aldrich, 1:1000) antibody or monoclonal mouse anti-β-actin (ZSGB-BIO, 1:1000) at 4 °C overnight. The membranes were then incubated with peroxidase-conjugated goat anti-rabbit and goat anti-mouse secondary antibodies (ZSGB-BIO, 1:3000) for 1 h. Then, the blots were visualized using the Quantitative Imaging System FluorChem™ SP (Thermo Fisher Scientific Inc.). The BDNF was normalized to β-actin bands.

2.6. The determination of monoamine neurotransmitter levels (Yi et al., 2012)

To explore the detailed neurochemical mechanisms involved in the antidepressant-like effect of albiflorin, the levels of hippocampal monoamine neurotransmitters and their metabolites were detected by HPLC with electrochemical detection (ECD) after the CUS protocol. The rats' tissues were homogenized in an ice-cold tissue lysis buffer containing 0.4 M $HClO_4$ (6.6 μL/mg), 0.5 mM Na_2EDTA and 0.1 g/L of L-cysteine. The samples were centrifuged at 15,000 × g for 15 min at 4 °C and then filtered through a 0.45 μm-pore membrane. The standard solution or sample was injected into the reversed-phase SunFire™ C18 column (5 μm, 250 mm × 4.6 mm). Separation was performed in an isocratic elution mode at a column temperature of 20 °C using a mobile phase containing 85 mM citric acid and 0.1 M sodium acetate buffer (pH 3.7) with 15% methanol, 0.9 mM sodium octanesulfonate, and 0.2 mM Na_2EDTA at a flow rate of 1.0 mL/min. The contents of the monoamine neurotransmitters and their metabolites were expressed as ng/g wet weight of tissue.

2.7. Statistical analysis

Statistical analysis was performed using GraphPad Prism (GraphPad Prism 5.0, version 2.0; GraphPad Software Inc., San Diego, CA). The results were expressed as the mean ± SEM.

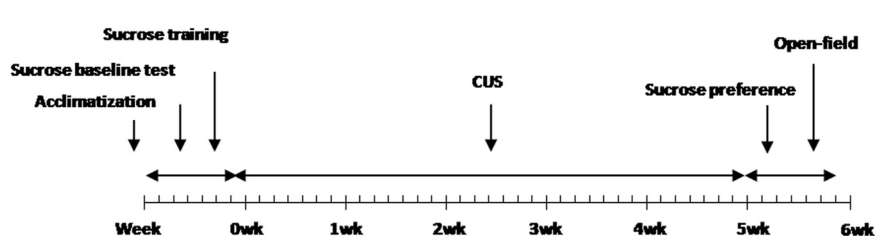

Fig. 2. The CUS procedure.

Student's t-test was used to compare the differences between two groups, while one-way analysis of variance (ANOVA) followed by Dunnett's test was used to compare the differences among three or more groups. A value of $P < 0.05$ was considered significant.

3. Results

3.1. Behavioral effects of subchronic treatment with albiflorin in mice

As shown in Fig. 3, subchronic treatment with fluoxetine (10 mg/kg, i.g., once per day for 7 d) produced an evident effect on immobility time in the FST ($F_{4,70}=4.855$, $P < 0.05$; Fig. 3A) and TST ($F_{4,70}=6.992$, $P < 0.001$; Fig. 3B). Further post-hoc analysis revealed that subchronic treatment with albiflorin at doses of 3.5, 7.0 and 14.0 mg/kg significantly decreased the immobility time in the FST and TST, indicating that albiflorin exerts an antidepressant-like effect in mice.

To dismiss the false positive results in the antidepressant effect, we observed the locomotor activity mice treated with albiflorin. As shown in Fig. 3C, we found that albiflorin had no significant effect in the general locomotor activity in mice across the dose range studied.

3.2. The results of the CUS procedure of rats

3.2.1. The effects of albiflorin on sucrose preference in CUS rats

Fig. 4 illustrates the effects of albiflorin on sucrose preference in CUS rats. The five-week stress procedure caused a significant decrease in sucrose preference in the stress-control rats compared with the control non-stressed rats (Student's t-test, $t=3.426$, $P < 0.01$). As a positive control, fluoxetine (10 mg/kg) significantly increased the sucrose preference in CUS rats ($F_{4,45}=3.850$, $P < 0.05$). Further post-hoc analysis revealed that albiflorin at 7.0 and 14.0 mg/kg restored the sucrose preference to normal levels ($P < 0.01$, compared with the stress-control group), indicating that albiflorin exerts an antidepressant-like effect in a CUS model of rats.

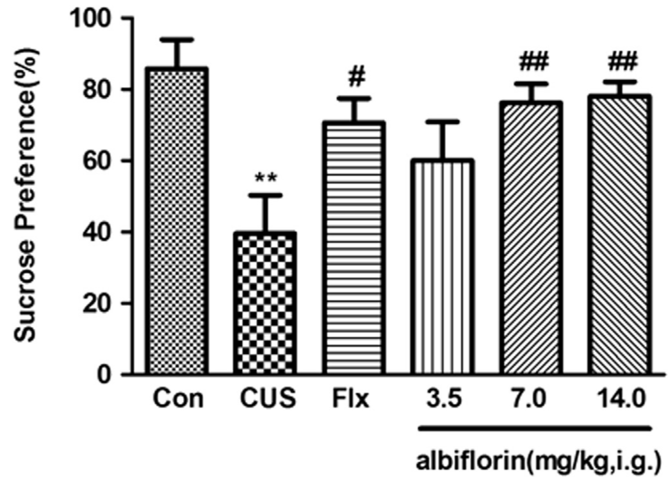

Fig. 4. The effects of albiflorin treatment on the sucrose preference in CUS rats. After 5 weeks of CUS, the sucrose preference was measured for a period of 1 h after 23 h of food and water deprivation. The data represent mean ± SEM, $n=10$. $^{**}P < 0.01$ vs. control; $^{\#}P < 0.05$; $^{\#\#}P < 0.01$ vs. CUS alone.

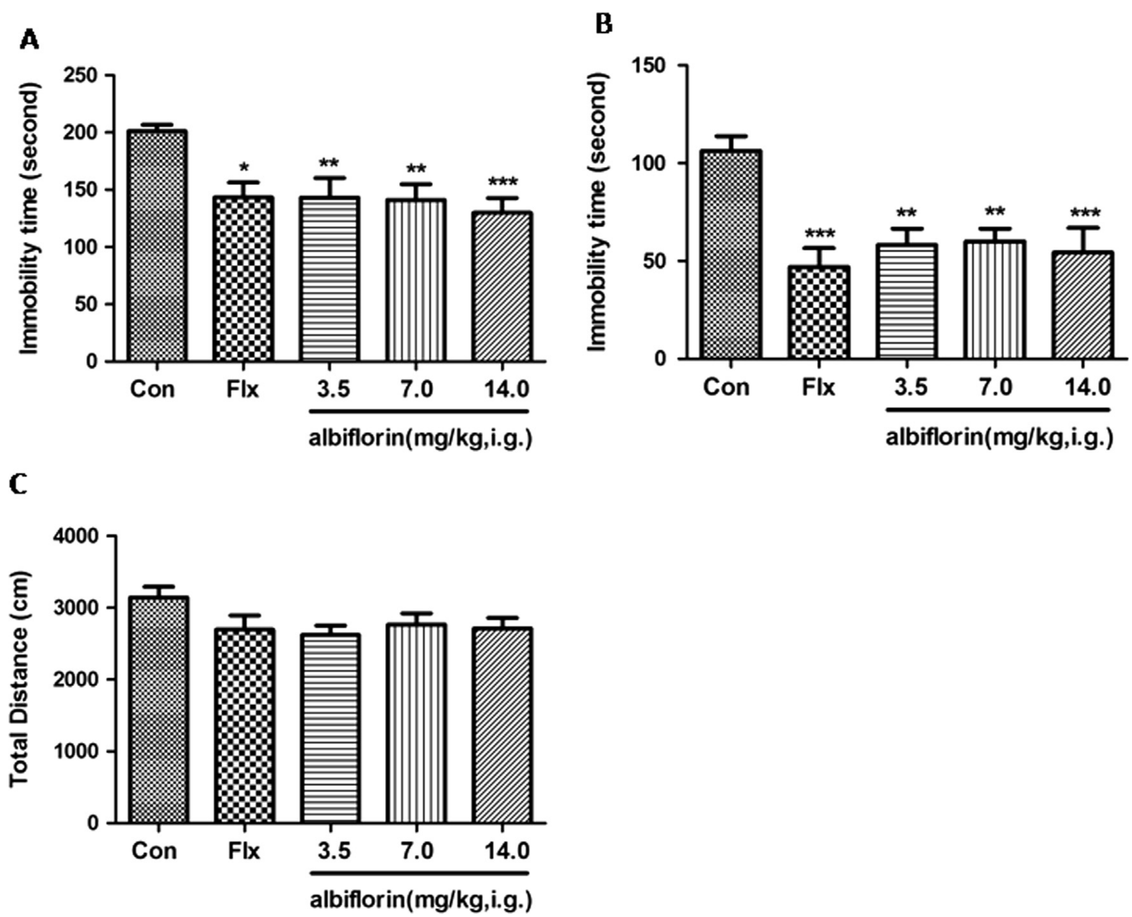

Fig. 3. The effect of albiflorin on the immobility time in FST (A) and TST (B) and locomotor activity (C) in mice. After oral administration for 7 days, the mice were subjected to behavioral tests. The data are the mean ± SEM, $n=15$. $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$ vs. control.

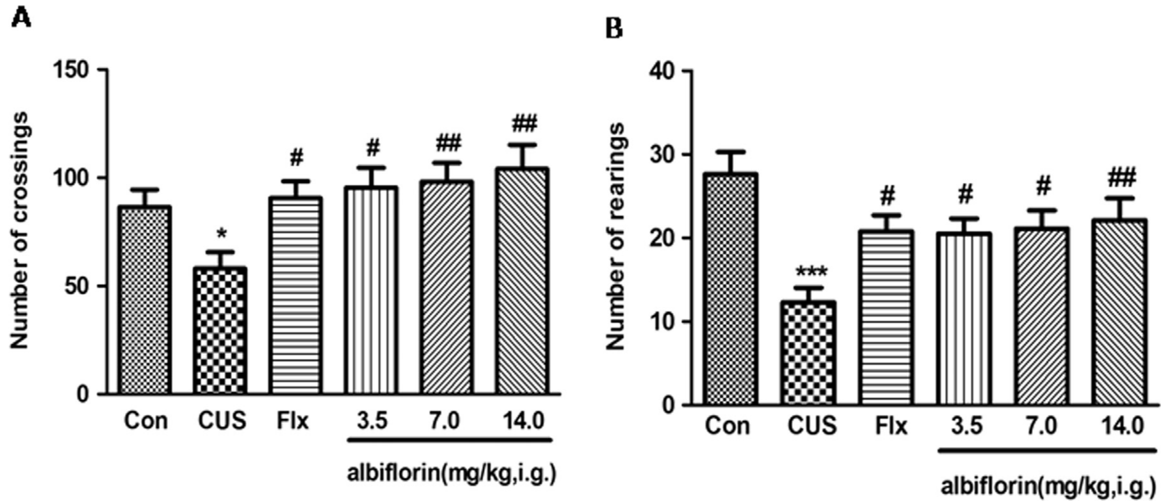

Fig. 5. The effects of chronic albiflorin treatment on the open-field behaviors of CUS rats. After 5 weeks of CUS, the number of crossings (A) and rearings (B) in 5 min was measured. The data represent mean ± SEM, n=10. $^{*}P<0.05$, $^{***}P<0.001$ vs. control; $^{\#}P<0.05$, $^{\#\#}P<0.01$ vs. CUS alone.

3.2.2. The effect of albiflorin on the open-field behaviors of CUS rats

Compared with the non-stressed control rats, the rats in the stress-control group displayed a significantly reduced number of crossings (Student's t-test, t=2.574, $P<0.05$) and rearings (Student's t-test, t=4.758, $P<0.001$) after a five-week exposure to CUS (Fig. 5). As a positive control, fluoxetine (10 mg/kg) significantly increased the number of crossings and rearings in CUS rats ($F_{4, 45}$=4.107, $P<0.05$ and $F_{4, 45}$=3.648, $P<0.05$, for 5A and 5B, respectively) compared with the stress-control group. Daily oral administration of albiflorin (3.5 mg/kg to 14.0 mg/kg) showed an evident increase in the number of crossings (Fig. 5A; $P<0.05$; $P<0.01$; $P<0.01$, 3.5 mg/kg, 7.0 mg/kg and 14.0 mg/kg, respectively) and rearings at 3.5 mg/kg to 14.0 mg/kg (Fig. 5B; $P<0.05$; $P<0.05$; $P<0.01$, respectively).These results further indicate the antidepressant-like effect of albiflorin in the CUS procedure.

3.3. The effects of albiflorin on BDNF expression

After 7 d of subchronic treatment with albiflorin in mice, a significant antidepressant-like effect was observed (shown in Fig. 3), and fluoxetine (10 mg/kg) ($F_{4,10}$=6.539, $P<0.05$; Fig. 6A) and albiflorin (7.0–14.0 mg/kg, $P<0.05$, $P<0.01$, $P<0.01$, respectively; Fig. 6A) significantly increased the hippocampal BDNF expression compared to the control group. In the CUS protocol, hippocampal BDNF levels were decreased in the hippocampus of stress-control rats (Student's t-test, t=4.501, $P<0.01$; Fig. 6B) while albiflorin (3.5–14.0 mg/kg) ($F_{4,15}$=5.350, $P<0.05$; $P<0.01$; $P<0.01$, respectively) or fluoxetine (10 mg/kg) ($P<0.05$) restored

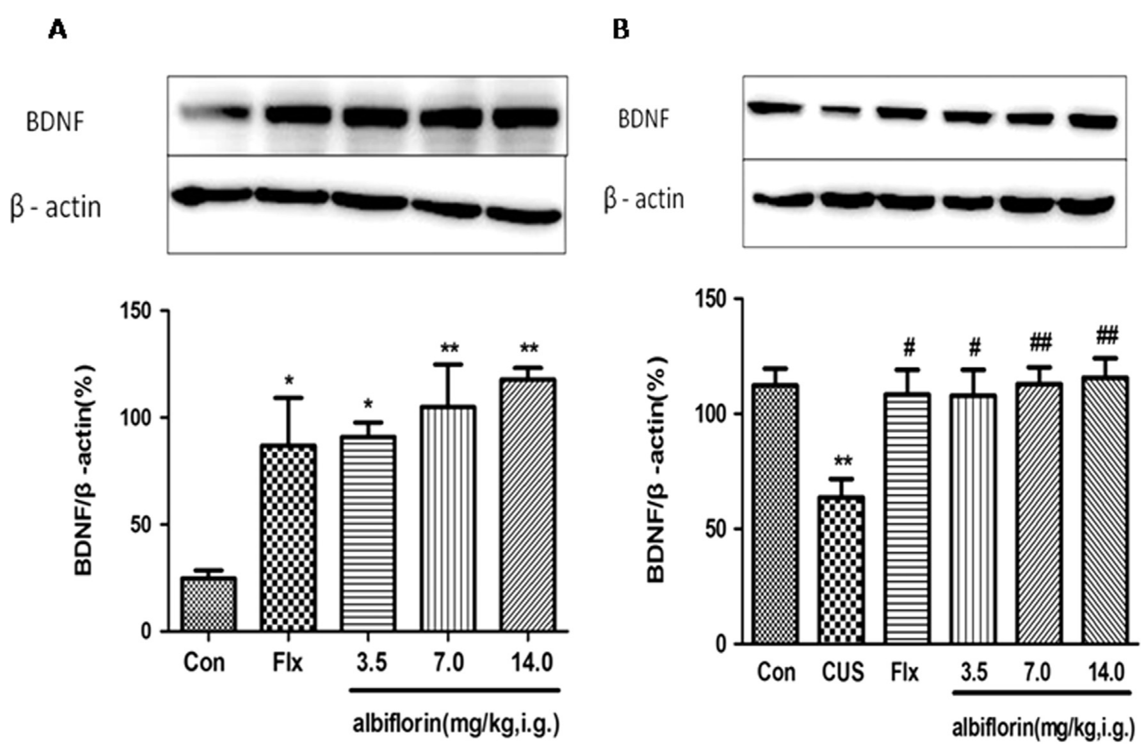

Fig. 6. The effects of repeated treatments of albiflorin on the BDNF expression in mice and chronically stressed rats. After treatment with albiflorin for 7 d in mice (A) or 37 d in chronically stressed rats (B), the hippocampal BDNF expression levels were measured by western blotting. The intensity of western bands was quantified with a densitometric scanner. Each column represents the mean ± SEM, n=3–4. $^{*}P<0.05$, $^{**}P<0.01$ vs. control; $^{\#}P<0.05$, $^{\#\#}P<0.01$ vs. CUS alone.

Table 1
The effects of albiflorin on the hippocampal monoamine neurotransmitter levels in CUS rats.

Group		5-HT	5-HIAA	NE	AD	HVA	DA	DOPAC
Control		38.1 ± 10.2	317.9 ± 41.0	562.3 ± 74.2	148.8 ± 18.5	7.1 ± 1.5	6.1 ± 2.0	13.6 ± 5.9
CUS		7.9 ± 0.7*	180.1 ± 15.0*	296.7 ± 32.3*	105.9 ± 8.0	4.4 ± 0.6	2.0 ± 0.5	22.4 ± 10.1
Fluoxetine	10 mg/kg	45.2 ± 11.0#	326.0 ± 46.5#	537.5 ± 36.19##	143.5 ± 17.0	8.1 ± 2.7	4.8 ± 2.3	12.6 ± 3.5
albiflorin	3.5 mg/kg	46.2 ± 12.5#	311.8 ± 28.3#	539.3 ± 63.9##	141.4 ± 11.6	13.8 ± 6.9	3.5 ± 1.7	12.7 ± 2.1
	7.0 mg/kg	47.6 ± 9.8#	275.3 ± 27.4	532.7 ± 54.9##	172.7 ± 24.7#	6.9 ± 1.9	8.0 ± 2.8	7.2 ± 1.4
	14.0 mg/kg	78.6 ± 9.7###	350.6 ± 45.2###	602.3 ± 46.0###	140.9 ± 10.1	6.0 ± 1.2	7.7 ± 1.8	10.6 ± 2.4

The data were expressed as the mean ± SEM in ng/g for six rats in each group.
DA=dopamine, NE=norepinephrine, AD=adrenalin, DOPAC=3,4-dihydroxyphenylacetic acid, HVA=homovanillic acid, 5-HT=5-hydroxytryptamine, 5-HIAA=5-hydroxyindoleacetic acid.
Hippocampal monoamine neurotransmitter levels were detected by HPLC with electrochemical detection (ECD) after the CUS protocol.

* $P < 0.05$ vs. the control group.
\# $P < 0.05$,
\## $P < 0.01$, and,
\### $P < 0.001$ vs. the CUS group.

the BDNF expression to normal levels (Fig. 6B), demonstrating a similar change using drugs and the antidepressant-like behaviors, indicating that the antidepressant-like effect of albiflorin may be closely related to a hippocampal BDNF increase.

3.4. *The effects of albiflorin on the hippocampal monoamine neurotransmitter levels in CUS rats*

The levels of monoamine neurotransmitters and their metabolites detected in the hippocampus are summarized in Table 1. The five-week stress procedure caused a significant decrease in 5-HT (Student's *t*-test, $t=2.944$, $P < 0.05$), 5-HIAA ($t=3.155$, $P < 0.05$), and NE ($t=2.836$, $P < 0.05$) levels in the stress-control rats compared with the control non-stressed rats. The 5-HT ($F_{4,\ 25}=6.733$, $P < 0.05$, $P < 0.05$, $P < 0.001$, 3.5–14.0 mg/kg, respectively), 5-HIAA ($F_{4,\ 25}=3.706$, $P < 0.05$, $P < 0.01$, 3.5 and 14.0 mg/kg, respectively), and NE ($F_{4,\ 25}=5.991$, $P < 0.01$, $P < 0.01$, $P < 0.001$,3.5–14.0 mg/kg, respectively) levels increased after albiflorin treatment. Fluoxetine (10 mg/kg) also induced elevations in 5-HT ($P < 0.05$), 5-HIAA ($P < 0.05$), and NE ($P < 0.01$) levels.

4. Discussion

Despite the wide use of *Radix paeoniae* Alba for treating mood disorders, there is a lack of scientific reports evaluating its pharmacological effects. In the present study, we first found that albiflorin, a main component of *Radix paeoniae* Alba, produced significant antidepressant-like effects in FST and TST in mice, without modifying the locomotor activity. We then found that albiflorin can also exert significant antidepressant-like effects in a rat CUS model of depression while increasing the hippocampal 5-HT/NE and BDNF levels. Albiflorin produced significant antidepressant-like effects closely related to the hippocampal 5-HT/NE increase and BDNF expression, which were similar to the classic antidepressant SSRIs.

FST and TST have been widely used to screen for antidepressant activity (Porsolt et al., 1977; Steru et al., 1985; Qiu et al., 2013a). In the tests, mice were under a stressful environment from which they cannot escape. Mice would become immobile after an initial period of struggling, which is similar to human depression and is feasible to reversal by antidepressant drugs. The anti-depressant effects of albiflorin were firstly evaluated by using these two models. It was observed that albiflorin significantly reduced the immobility time in these two models, and the decrease in immobility time was dose dependent, which is consistent with previous experiments. To avoid the false-positive results (Bourin et al., 1983; Zheng et al., 2013), the effect of albiflorin on locomotor activity was evaluated using the locomotor activity test. The results showed that albiflorin did not increase the locomotor activity, suggesting that albiflorin has an antidepressant-like effect in FST and TST. Albiflorin at a dose of 14.0 mg/kg was more efficacious than that of 3.5 and 7.0 mg/kg.

CUS is a wildly used animal model of depression where animals are exposed to a combination of mild unpredictable stressors (Willner, 1997). Many animal studies have shown that CUS can induce behavioral changes that resemble clinical depression, such as reduced sucrose intake, altered weight gain, locomotor activity deficit, degraded physical state, and decreased responsiveness to rewarding stimuli (Duncko et al., 2001; Gronli et al., 2005; Papp et al., 1993). These CUS-induced behavioral parameters can be blocked or reversed by the chronic administration of antidepressants (Willner, 2005). The sucrose preference test is an index of anhedonia-like behavioral changes. Anhedonia, a core symptom of major depressive disorder, is modeled by inducing a generalized decrease in responsiveness to rewards, which is reflected by a reduced intake of palatable sweet solutions (Willner et al., 1987). Our results showed that the albiflorin group at 7.0 and 14.0 mg/kg restored the sucrose preference to normal levels compared with the CUS group, suggesting that albiflorin has antidepressant-like effects. In this model, the effects of stress were assessed by changes in open field activity, which is used in response to acute stress such as noise, and the activity was decreased (Vollmayr and Henn, 2003). Our results found that a reduction in open field activity induced by the CUS procedure was significantly reversed by chronic treatment with albiflorin. Current tests also demonstrated that oral administration of 3.5–14.0 mg/kg albiflorin was the available dosage range, which is consistent with the results of the FST and TST.

As the most classic pathophysiological theory of depression, the monoamine hypothesis holds that low levels of serotonin and norepinephrine, these substances in the central nervous system will be closely related to depression (Delgado, 2000). Substantial evidence obtained from clinical and preclinical studies indicated that mental depression was due to an absence of brain monoaminergic activity (Huang et al., 2012), and the levels of monoamine neurotransmitters in the brain increased in the experimental groups compared with those in the control group after antidepressant treatments (Elhwuegi, 2004). BDNF, known as the endogenous neurotrophic factor that modulates neuronal plasticity and inhibits cell death cascades, is the best studied factor in depression (Chourbaji et al., 2011; Castren et al., 2007; Voleti and Duman, 2012). BDNF has one effect relevant to the therapeutic actions of antidepressants: the regulation of serotonin neurons; local infusion of BDNF into the cerebral cortex or midbrain elevates serotonin levels (Siuciak et al., 1994) and protects serotonin neurons from neurotoxin-induced damage (Mamounas et al., 1995).

In our study, CUS decreased the levels of 5-HT, 5-HIAA and NE in the hippocampus, and these effects were normalized by albiflorin treatment. The expression levels of hippocampal BDNF are up-regulated by sub-chronic and chronic albiflorin treatment in both naive and stressed animals at all of the doses. These findings suggest that albiflorin might generate antidepressant-like effects via increasing the hippocampal 5-HT/NE and BDNF levels.

5. Conclusion

In summary, albiflorin displayed an antidepressant-like effect in mice and rats, and this effect may be partially mediated by regulating the central monoamine neurotransmitter system and increasing the levels of BDNF, which is similar to the effects of fluoxetine. However, the results of this study would benefit from further investigation into the antidepressant mechanism of this potential treatment option.

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